

Grade 6 Accelerated
Unit 3
Lesson 9 Homework

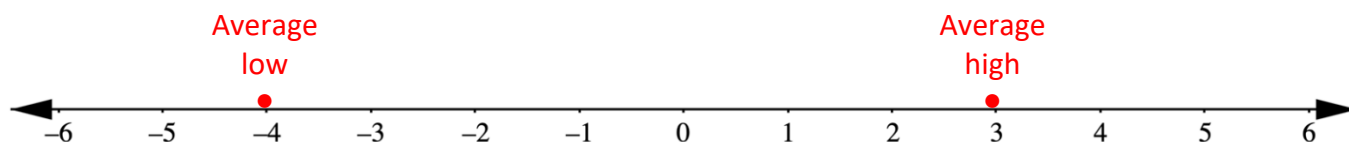
Name Answer Key
Date _____
Period _____

The freezing point is 0°C . In December in Chicago, Illinois, the average high temperature is 3°C above freezing and the average low temperature is 4°C below freezing.

1. Write the values that represent the average high temperature and the average low temperature in Chicago in December.

Description	Value
Average high temperature	3
Average low temperature	-4

2. Model the two values on the number line.



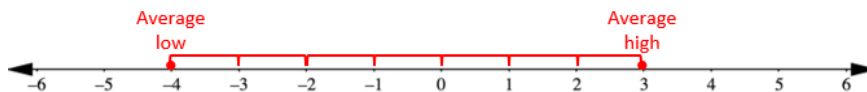
3. Write a mathematical statement using absolute value that could be used to find the difference between the average high and average low temperatures.

$$|-4| + |3|$$

4. What is the difference between the average high temperature and the average low temperature?

$$|-4| + |3|$$

$$4 + 3 = 7$$



5. Write a mathematical statement using absolute value that could be used to determine how much farther from freezing the average low temperature is than the average high temperature.

$$|-4| - |3|$$

6. How much farther from freezing is the low temperature than the high temperature?

$$|-4| - |3|$$

$$4 - 3 = 1$$

Allan withdrew money from his bank account three times last week. Each time, the transaction on his bank statement said $-\$40$.

7. Write a mathematical statement using absolute value that could be used to determine the total amount of money Allan withdrew last week.

$$3 \times |-40|$$

8. What is the total amount of money Allan withdrew last week?

$$\begin{aligned} 3 \times |-40| & \quad \text{Allan withdrew \$120 last week.} \\ 3 \times 40 & \\ = 120 & \end{aligned}$$

A dolphin and a turtle are both swimming. The dolphin is 6 feet below sea level, and the turtle is 31 feet below sea level.

9. Write a mathematical statement using absolute value that could be used to determine how much farther from sea level the turtle is than the dolphin.

$$|-31| - |-6|$$

10. How much farther from sea level is the turtle than the dolphin?

$$\begin{aligned} & |-31| - |-6| \\ & 31 - 6 \\ & = 25 \end{aligned}$$

The turtle is 25 ft further below sea level than the dolphin is.